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PERSPECTIVE



Cause and effect? A review of the impact of antibiotics on the gut microbiome in patients undergoing hematopoietic stem cell transplantation

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ABSTRACT

Introduction: Patients undergoing allogeneic hematopoietic stem cell transplantation (allo-HSCT) are at high risk of infection due to immunosuppression, prompting routine use of prophylactic and broad-spectrum antibiotics for treatment. However, emerging evidence suggests that gut microbiome disruption (dysbiosis), partly caused by antibiotic use, is associated with poorer transplant outcomes, including graft-versus-host disease (GVHD), infection, and mortality.

Areas covered: This narrative review discusses antibiotic use to prevent and treat febrile neutropenia in allo-HSCT recipients, including effectiveness, impacts on microbiome and GVHD, antimicrobial resistance and *Clostridioides difficile* infection (CDI). It also reviews available strategies to reduce unnecessary antibiotic use and proposes potential future interventions. A comprehensive PubMed search was conducted through 2024 using terms related to HSCT, antimicrobials, microbiome, resistance, and CDI.

Expert opinion: Improving outcomes while minimizing emergence of antibiotic resistance and CDI requires personalized, risk-adaptive antimicrobial stewardship (AMS). Tailored AMS approaches, including patient risk stratification and early de-escalation, could limit unnecessary antibiotic use and mitigate adverse effects. In the future, microbiome preservation and restoration may reduce transplant complications by maintaining colonization resistance and immune balance. Integrating these strategies into allo-HSCT care is essential for optimizing both clinical and microbiological outcomes.

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Hematopoietic stem cell transplantation; antibiotics; graft-versus-host disease; microbiome; beta-lactams; fluoroquinolones; glycopeptides

1. Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HSCT), or bone marrow transplantation, is a curative therapy for high-risk hematological malignancies including leukemia and myelodysplastic syndromes. Patients undergoing allo-HSCT are highly susceptible to infections due to cytopenias caused by the underlying hematological malignancy, conditioning therapies and immunosuppression given to control off target effects of the allo-HSCT graft, including graft-versus-host disease (GVHD). To mitigate this risk, antimicrobials, including antibiotics, antivirals and antifungals, are routinely administered. In allo-HSCT, antibiotics are used for prophylaxis: typically fluoroquinolones, empirical treatment of febrile neutropenia (FN) of unknown origin: typically beta-lactams, with or without additional gram-positive coverage, and targeted therapy when specific pathogens are identified.

FN is observed in the majority of allo-HSCT recipients and associated with significant morbidity and mortality [1,2]. It is well recognized that the time to antibiotic administration is a critical factor in reducing mortality [3]. Therefore, once high-risk patients develop FN, immediate treatment with intravenous broad-spectrum antibiotics is essential. While antibiotics are essential and life-saving treatments, extensive use is

associated with the emergence of antimicrobial resistance (AMR) through selection of multi-drug resistant (MDR) organisms. MDR pathogens can lead to clinical infections that are difficult or impossible to treat at a patient level, as well as adverse consequences for infection control at an institutional level and broader implications for public health.

Indeed, AMR is a major global health concern. In 2021, an estimated 4.71 million deaths were associated with bacterial AMR, with 1.14 million deaths directly attributable to resistant infections [4]. Projections suggest that by 2050, AMR could be associated with 8.22 million deaths annually, with 1.91 million directly attributable to MDR infections. Additionally, antibiotic use is linked to unintended toxicities, including *Clostridioides difficile* infection (CDI) [5], increased susceptibility to certain childhood diseases following early antibiotic exposure [6], and reduced efficacy of cancer immunotherapies [7,8]. Many of these adverse effects are believed to stem from disruptions to the gut microbiome.

The gut microbiome is now recognized to play an important role in human health and disease. This complex ecosystem, composed of trillions of microorganisms, plays a crucial role in development of immune tolerance [9], digesting

Article highlights

- Disruption of the gastrointestinal microbiome is associated with increased risk of graft-versus-host disease (GVHD) and mortality after allo-HSCT.
- Broad-spectrum antibiotics, particularly those with anaerobic coverage, significantly impact the microbiome and negatively influence allo-HSCT outcomes.
- Fluoroquinolone prophylaxis and additional gram-positive agents appear to have less effect on the microbiome and GVHD; however, due to limited evidence of clear benefits and growing antimicrobial resistance (AMR), their use should be limited.
- Personalized, risk-adapted antibiotic strategies and avoiding over-treatment may reduce antibiotic-induced dysbiosis, AMR, and super-infection risks.
- Future antibiotic development and prescribing practices should account for their potential impact on the microbiome.

complex carbohydrates [10], synthesizing vitamins [11], producing short-chain fatty acids and maintain gut barrier integrity [12]. Moreover, the gut microbiome modulates immune function and development [13], prevents pathogen colonization [14] and influences brain function through the gut-brain axis [15]. In the last decade, the association between a disrupted gut microbiome and adverse allo-HSCT outcomes have been extensively studied. In an early study, Jenq *et al.* found that a loss of overall microbiome diversity, a decrease in *Clostridiales*, and an increased abundance of *Lactobacillales* was associated with a higher risk of GVHD in both animal models and human patients [16]. This finding suggested the potential for microbiome-targeted therapies in allo-HSCT. Taur *et al.* demonstrated that patients with higher microbiome diversity had improved three-year overall survival, with fewer infections and lower GVHD incidence [17]. These survivors had a greater abundance of *Lachnospiraceae* and *Actinomycetaceae*, while those with low microbiome diversity were dominated by single genera such as *Enterococci*, *Streptococci*, *Enterobacteriaceae*, and *Lactobacilli*. Further supporting these findings, Weber *et al.* used urinary 3-indoxyl sulfate as a biomarker for microbiome disruption and found that low 3-indoxyl sulfate levels were significantly associated with increased transplant-related mortality [18]. In a large international multicenter observational study, Peled *et al.* reported that lower microbiome diversity before and during allo-HSCT was associated with an increased risk of GVHD and reduced survival, but only in patients receiving T-cell replete allo-HSCT [19]. The microbiota affecting clinical outcomes are summarized in a review by Shono *et al.* [20].

A comprehensive PubMed search was conducted through 2024, using 'HSCT,' 'antibiotics,' 'microbiome' and their synonyms and related terms. Additional search was conducted using 'CDI,' 'resistance' and their synonyms excluding 'microbiome,' for the specific topics. References of selected papers were also reviewed. In total, 482 papers were screened and 145 were included in this review. This review examines the impact of antibiotics commonly used in allo-HSCT recipients for the treatment and prophylaxis of FN on microbiome-related clinical outcomes and provides insights into potential practice improvements to optimize patient care while mitigating associated risks. This review does not include antibiotic

uses not related to FN or prophylaxis, or microbiome-targeted interventions not related to antibiotics. Microbiome-targeted interventions are summarized in other reviews [20–25].

2. Broad-spectrum antibiotics for FN

2.1. Use of broad-spectrum antibiotics in FN

Prompt empiric treatment of FN with broad-spectrum intravenous antibiotics is associated with reduced mortality [3]. The choice of empirical antibiotics depends on local epidemiology and resistance patterns as well as patient colonization with AMR organisms. Coverage of gram-negative bacteria and in particular *Pseudomonas aeruginosa* is necessary. As a first-line treatment, an anti-pseudomonal cephalosporin such as cefepime, or beta-lactam-beta-lactamase combination such as piperacillin-tazobactam is used [2]. In settings with a high prevalence of extended-spectrum beta-lactamase (ESBL)-producing gram-negative bacteria, carbapenems such as meropenem may be necessary. These agents should be administered as monotherapy. In patients experiencing septic shock, there is a role for dual therapy with a beta-lactam and an aminoglycoside, which is associated with reduced mortality [26]. Fluoroquinolones should be avoided as treatment for FN, especially if they were used as prophylaxis. Comparative studies suggest that cefepime, piperacillin-tazobactam and carbapenems have similar efficacy and safety profiles [27–29].

2.2. Impact of broad-spectrum antibiotics on GVHD and mortality

Numerous studies have examined the impact of broad-spectrum antibiotics on allo-HSCT outcomes, particularly in relation to GVHD and mortality (Table 1). Shono *et al.* analyzed 857 patients, finding that FN treatment with imipenem-cilastatin or piperacillin-tazobactam increased GVHD and mortality, whereas aztreonam and cefepime did not [30]. Murine models confirmed that piperacillin-tazobactam and imipenem significantly depleted *Clostridiales*, a key bacterial group, while cefepime and aztreonam preserved them [30]. A larger retrospective study of 2,023 allo-HSCT recipients by Rashidi *et al.* linked post-transplant exposure to carbapenems or piperacillin-tazobactam to increased GVHD risk [31]. Some prospective studies have examined microbiome shifts caused by different antibiotics. Morjaria *et al.* and Norgaard *et al.* demonstrated that piperacillin-tazobactam and meropenem significantly reduced microbiome diversity and depleted anaerobic bacteria [32,33]. These results suggest that antibiotics with anaerobic coverage contribute to GVHD development. A Phase II trial (NCT03078010) is currently comparing piperacillin-tazobactam and cefepime to assess their effects on the microbiome and transplant outcomes. In pediatric patients ($N=2,550$), carbapenems were associated with a higher risk of grade 2–4 GVHD, while penicillins and cephalosporins showed no significant association [34]. Additional smaller retrospective studies and meta-analyses have reported similar findings, reinforcing concerns about anti-anaerobic antibiotics increasing GVHD risk [35–39].

The timing of antibiotic administration also plays a role. A retrospective analysis of 621 allo-HSCT categorized patients into three groups: antibiotics before allo-HSCT (Early ABs), after allo-HSCT (Late ABs), or no antibiotics (No ABs) [40]. Treatment-related mortality (TRM) varied significantly across these groups: 34% in the Early ABs group, 21% in the Late ABs group, and only 7% in the No ABs group. Clostridial abundance was highest in the No ABs group, lower in the Late ABs group, and nearly absent in the Early ABs group [40]. Additional references are shown in Table 1 [41–46].

Collectively, these findings highlight the complex interplay between broad-spectrum antibiotics, the gut microbiome,

and transplant outcomes in allo-HSCT recipients. While antibiotics remain essential for the treatment of infection, those with anaerobic activity, such as piperacillin-tazobactam and carbapenems, appear to increase GVHD and mortality risk. In contrast, non-anaerobic agents like cefepime may be safer alternatives. Of note, there are clinical situations in which anaerobe coverage should be prioritized, such as in cases of severe mucositis, neutropenic enterocolitis, suspected sepsis with a gut-derived organism, or systemic compromise. Further studies, including ongoing clinical trials, are necessary to refine antibiotic strategies and improve allo-HSCT outcomes.

Table 1. Studies on effects of broad-spectrum antibiotics on microbiome and GVHD/mortality.

Author	Country	Study design population	N	Clinical Outcomes	Gut microbiome	Other markers	Findings
Shono <i>et al.</i> [30]	USA	Retrospective cohort Adults	857	GVHD related mortality (5 years)	Y	N	Outcomes treated vs not treated imipenem-cilastatin: 21.5 vs 13.1% Piptaz: 19.8 vs 11.9% Aztreonam: 17.5 vs 14.2% (NS) Cefepime: 13.8 vs 14.6% (NS)
Rashidi <i>et al.</i> [31]	USA	Retrospective cohort Adults	2023	G2–4 aGVHD	N	N	Carbapenems: HR 2.75 Piptaz: HR6.55
Elgarten <i>et al.</i> [34]	USA	Retrospective Pediatric International registry data	2550	G2–4 aGVHD	N	N	Carbapenems: aOR 1.24 ($p = 0.035$) Penicillins: aOR 1.24 ($p = 0.14$) Cephalosporins: aOR 1.05 ($p = 0.71$)
Hanks <i>et al.</i> [36]	USA	Retrospective Adults	273	G2–4 aGVHD	N	N	Anti-anaerobic ABs vs other ABs :OR = 2.3 CI = 1.1–4.8, $p = 0.021$
Hidaka <i>et al.</i> [37]	Japan	Retrospective cohort, Adults	200	Intestinal GVHD	N	N	Carbapenem increased GVHD ($p = 0.012$) Glycopeptide, penicillins and quinolones: NS
Holler <i>et al.</i> [41]	Germany	Prospective analysis of microbiome and 3-IS, Adults	31	GVHD and microbiome	Y	3-IS	Microbiome shifted to enterococcus and reduced 3-IS after AB (not specified) or with GVHD
Lee <i>et al.</i> [42]	Korea	Retrospective Adults	131	aGVHD, cGVHD and pre-HSCT AB use	N	N	aGVHD: NS (Piptaz, Cefepime, Carbapenems, Glycopeptide) cGVHD: associated with pre-HSCT glycopeptide
Lee SE <i>et al.</i> [38]	Korea	Retrospective Adults	211	Intestinal GVHD (treated vs not treated)	Y	Immune cells	Cefepime: NS (vs no ABs) Carbapenem: HR 3.25, CI 1.13–9.34, $p = 0.029$
Meedt <i>et al.</i> [43]	Germany	Prospective Adults	201	BCoAT, GVHD	Y	BCoAT 3-IS	Low BCoAT was associated with early AB use (before D0) and intestinal GVHD. AB not specified
Morjaria <i>et al.</i> [32]	USA	Prospective stool sample analysis Adults	18	Loss of obligate anaerobes by ABs	Y (daily sampling)	Modeling	Piptaz, and meropenem: >99% loss of obligate anaerobic bacteria
Nishi <i>et al.</i> [44]	Japan	Retrospective adults	241	G2–4 GVHD	N	N	4 th generation cephalosporins: HR 1.98, CI 1.19–3.29; $p = .0087$ Quinolones, Piptaz, carbapenems: NS
Norgaard <i>et al.</i> [33]	Denmark	Prospective Adults	233	Diversity and resistant gene	Y	N	Piptaz, meropenem: most affected to diversity No clinical outcomes
Taur <i>et al.</i> [17]	USA	Prospective stool sample analysis Adults	80	Diversity and mortality	Y	N	High diversity at engraftment was associated with high survival rate. Beta-lactam was used more frequently in patients with low diversity.
Weber <i>et al.</i> [18]	Germany	Prospective Adults	131	3-IS, diversity and GVHD	Y	3-IS	Low 3-IS level is associated with dysbiosis and poor survival, as well as early AB use
Waber <i>et al.</i> [46]	Germany	Retrospective Adults	394	3-IS, survival, enterococcal load	Y/N	3-IS	Rifaximin group had higher 3-IS, and lower TRM and enterococcal load compared to ciprofloxacin/ metronidazole group
Weber <i>et al.</i> [40]	Germany USA	Retrospective Adults	621	GVHD related TRM Microbiome composition 3-IS	Y (U.S.A.) N (Germany)	3-IS	ABs pre-HSCT vs post-HSCT vs no ABs TRM: 34 vs 21 vs 7% ABs: Piptaz first line, carbapenems second line
Weber <i>et al.</i> [39]	Germany	Retrospective Adults	161	Survival 3-IS	N	3-IS	3-IS and survival: Rifaximin only> Rifaximin + ABs> cipro/metro
Weber <i>et al.</i> [45]	Germany	Retrospective Adults	188	Days on ABs, Diversity, 3-IS and GVHD	Y	3-IS	Permissive vs restrictive use of ABs 3-IS & diversity was higher with restrictive ABs. Severe gut GVHD reduced with restrictive ABs

" >aGVHD: acute GVHD, cGVHD: chronic GVHD, G2–4: Grade 2–4, TRM: treatment related mortality, 3-IS: 3-Indoxyl Sulfate, ABs: antibiotics, BCoAT: Butyryl-coenzyme A, Piptaz: piperacillin-tazobactam.

2.3. Possible mechanisms for antibiotics increase GVHD

One of the key mechanisms linking microbiome disruption to GVHD involves short-chain fatty acids, which are major metabolic products of certain commensal bacteria [47]. In particular, butyrate plays a protective role against acute GVHD by improving intestinal epithelial cell (IEC) junctional integrity and regulating colonic regulatory T cells [12,48]. Payen *et al.* demonstrated that butyrate levels in stool samples decreased by 86% even in mild GVHD cases, while acetate and propionate reductions were more pronounced in severe GVHD cases [49]. In addition, recent pre-clinical studies suggest that microbiome may control major histocompatibility complex (MHC) class II expression on IECs, which enhances donor CD4+ T cell activation and exacerbates GVHD [50,51]. Other pre-clinical studies found that meropenem treatment upregulated *in vivo* expression of mucus-degrading enzymes by *Bacteroides thetaiotaomicron*, leading to thinning of the colonic mucus layer [52]. Supplementing with xylose significantly prevented colonic mucus thinning in meropenem-treated mice. Further research by the same group found that *B. ovatus* helped to reduce GVHD by suppressing mucus-degrading gut microbes by producing xylose [53].

2.4. Antimicrobial resistance and *C.difficile* infection (CDI).

2.4.1. Antimicrobial resistance and gut colonization

The growing threat of antimicrobial resistance (AMR) has significantly compromised the effectiveness of antibiotics. The widespread use of broad-spectrum antibiotics, particularly in immunocompromised populations such as allo-HSCT recipients, has accelerated the emergence of MDR organisms, including extended-spectrum beta-lactamase (ESBL)-producing Enterobacterales, carbapenemase-producing Enterobacterales (CPE) and vancomycin-resistant Enterococci (VRE) [54]. The gastrointestinal (GI) tract serves as a major reservoir for AMR organisms [55,56]. These resistant bacteria can colonize the lower GI tract, (i.e. without causing symptoms). However, when the integrity of the mucosal barrier is compromised – such as in allo-HSCT recipients – these pathogens can translocate into the bloodstream, leading to bacteremia that are difficult to treat [57]. Sequencing studies have shown that bacterial strains isolated from the bloodstream of patients undergoing HSCT (both allogeneic and autologous) were identical to those found in their gut microbiota, underscoring the role of mucosal barrier integrity and the gut as a primary source of MDR infections [57]. The spread of resistance is further facilitated by plasmid-mediated horizontal gene transfer, allowing beta-lactamase enzymes to be shared between bacterial populations [58]. A European study demonstrated that gut colonization by MDR bacteria in allo-HSCT recipients was associated with an increased risk of bacteremia, higher rates of grade 2–4 acute GVHD, and reduced overall survival [59].

2.4.2. Broad-spectrum antibiotics and antimicrobial resistance

A healthy gut microbiome plays a crucial role in protecting against pathogen colonization, a phenomenon known as colonization resistance [60]. The administration of broad-spectrum

antibiotics disrupts these protective mechanisms, leading to microbiome dysbiosis, a loss of colonization resistance, and an increased susceptibility to infections, including those caused by MDR organisms [61]. The peri-transplant use of carbapenems and colistin has been particularly associated with MDR gram-negative bacilli bacteremia and a higher incidence of GVHD in allo-HSCT recipients [62]. A large international retrospective cohort study also demonstrated that prior therapy with piperacillin-tazobactam, antipseudomonal carbapenems, and fluoroquinolone prophylaxis are risk factors for blood stream infections due to MDR *Pseudomonas aeruginosa* in cancer patients [63].

A recent pediatric study analyzed the impact of anaerobic-versus aerobic-targeting antibiotics on the prevalence of antimicrobial resistance genes [64]. The study found that antibiotics with anaerobic coverage, such as piperacillin-tazobactam, carbapenems, metronidazole, and clindamycin, were linked to a higher abundance of antimicrobial resistance genes compared to antibiotics targeting aerobic bacteria, including cefepime, vancomycin, fluoroquinolones, aminoglycosides, macrolides, and cotrimoxazole. Several studies have demonstrated the effectiveness of fecal microbiota transplantation (FMT) in reducing or decolonizing multidrug-resistant organisms [65,66].

2.4.3. Broad-spectrum antibiotics and CDI in HSCT

CDI is a major complication of antibiotic use in patients undergoing autologous or allogeneic HSCT. The loss of colonization resistance due to dysbiosis, induced partly by antibiotics, allows *C. difficile* to proliferate, producing toxins that cause colitis and severe diarrhea. The incidence of CDI in HSCT is reported 9–31%, depending on the observation period and HSCT types in the studies [67–74]. CDI usually occurs early phase of HSCT (within a month), and most cases are mild. Reported risk factors include use of broad-spectrum antibiotics [67], 3rd and 4th generation cephalosporins [70], vancomycin use and VRE colonization [75], and steroid use for GVHD [68]. While some studies have suggested CDI as a risk factor of subsequent GVHD [67,68,70,71], other studies suggested GVHD being a risk factor for CDI [67,74]. Of note, many HSCT recipients also take proton pump inhibitors, which is another risk factor for CDI [76].

2.5. Pharmacokinetics and microbiome

Most intravenous broad-spectrum antibiotics are partly excreted in the bile and reach lower gastrointestinal tract, where they may cause dysbiosis. A culture-based study compared the fecal concentrations of intravenous antibiotics and their effects on the fecal microbiome [77]. Cefoperazone, which is primarily eliminated via bile and reaches the highest fecal concentrations, had the most significant impact on enterococci, anaerobes, and yeasts. In contrast, piperacillin-tazobactam was almost undetectable in fecal samples in this study. However, more recent studies suggest that renal excretion accounts for only 50–60% of piperacillin-tazobactam elimination, with high concentrations also detected in bile despite biliary excretion reportedly being less than 5% [78,79]. Given

the high gastrointestinal concentration of piperacillin-tazobactam, gastrointestinal secretion may partly explain its detrimental effects on the gut microbiome [78]. Ideally, intravenous antibiotics that are fully eliminated via the kidneys and not excreted into the gut would be preferred. However, such antibiotics are rare, with aminoglycosides being one of the few examples. There are currently few studies assessing the impact of different routes or methods of antibiotic administration on the gut microbiome. Finally, dysbiosis can also influence the pharmacokinetics of various drugs raising additional healthcare decision-making considerations. Emerging evidence suggests that the gastrointestinal microbiome can affect the pharmacokinetics (PK) of certain drugs, including commonly used drugs in allo-HSCT, such as oral tacrolimus [80] and intravenous mycophenolate [81].

2.6. Practice points to reduce antibiotic exposure

In the treatment of FN in patients with cancer, the selection of either a piperacillin-tazobactam or 4th generation cephalosporins is driven by regional practices. Except retrospective observations in patients undergoing allo-HSCT, there is no clear survival benefit with one over the other. Currently cefepime is the most commonly used first-line empirical antibiotic for FN in the United States [82], whereas piperacillin-tazobactam remains the predominant choice in Europe [83].

If using broad-spectrum antibiotics with anaerobe cover, one possible approach is the restrictive use of antibiotics with early de-escalation. A Spanish randomized trial demonstrated that discontinuation of empirical antibiotics after ≥ 72 hours in afebrile and clinically stable patients did not increase recurrent fever rates (14% in study group vs 18% in control), suggesting this approach is safe [84]. Similarly, a U.S. retrospective cohort study found that antibiotic de-escalation did not lead to an increase in recurrent fever (15% vs 19% in control), length of hospital stay (20 vs 20 days), ICU admissions (0 vs 3%), or CDI (4 vs 1%) [85]. Weber *et al.* evaluated the impact of transitioning from a permissive to a restrictive antibiotic use strategy [45]. In this retrospective cohort study, 'restrictive' use was defined as delaying antibiotic initiation when fever occurred after the infusion of antithymocyte globulin (ATG) or blood products. This approach led to a reduction in antibiotic exposure (duration 15.9 vs 21.7 days), increased microbiome diversity (Simpson and Shannon Index), and showed a trend toward decreased grade 2–4 acute GVHD (8.0% vs 15.8%, $p=.1$). To support the restrictive use of broad-spectrum antibiotics, distinguishing noninfectious fever causes, such as cytokine release syndrome or mucositis, is essential [86,87]. Additionally, rapid molecular identification of infectious pathogens would allow for more targeted antibiotic use, minimizing unnecessary broad-spectrum antibiotic exposure [88].

Another approach is active removal of antibiotics from the colon. Several experimental approaches aim to minimize antibiotic concentrations in the colon while preserving systemic efficacy. One such strategy is to use an orally administered beta-lactamase, ribaxamase that degrades ceftriaxone in the gut after IV administration. Studies have shown that ribaxamase reduces the acquisition of AMR genes and gene variants,

including those encoding β -lactamases and vancomycin resistance [89]. Another strategy involves the use of activated charcoal, which is already widely used to treat oral overdose by binding drugs in the gut. Activated charcoal has been explored as a means to protect the gut microbiome from antibiotic exposure. De Gunzburg *et al.* developed a colon-targeted activated charcoal formulation (DAV132) with a time- and pH-dependent drug delivery system to reduce interference with systemic drug absorption [90]. In a study, DAV132 co-administered with oral moxifloxacin preserved microbiome diversity without reducing plasma moxifloxacin concentrations [91]. However, the practical application of activated charcoal in allo-HSCT patients may be limited by the large quantities required for antibiotic absorption [92].

3. Additional gram-positive agents

3.1. The role of additional gram-positive agents

In addition to broad-spectrum beta-lactam antibiotics, the use of additional gram-positive agents such as glycopeptide, linezolid, or daptomycin has been common practice, and there has been a shift from gram-negative to gram-positive infections in FN [93]. However, there is no established evidence supporting the routine use of these agents. Cochrane reviews have found no clear benefits of the nonselective empirical use of glycopeptides, either at the onset of FN or for persistent fevers [94,95]. There remains inadequate trial data to identify any specific patient subgroups that might benefit from additional gram-positive coverage. The lack of a demonstrated mortality benefit may be explained by the fact that the majority of infection-related deaths in neutropenic patients are associated with gram-negative rather than gram-positive organisms [26,96,97].

The Infectious Diseases Society of America (IDSA) guidelines recommend against the routine use of vancomycin or other agents targeting drug-resistant aerobic gram-positive cocci. However, they outline specific clinical scenarios where gram-positive coverage may be warranted, including severe sepsis, pneumonia, positive blood cultures for gram-positive bacteria, catheter-related infections, skin or soft tissue infections, and colonization with antibiotic-resistant organisms [98]. Similarly, European Society for Medical Oncology (ESMO) guidelines recommend vancomycin or alternative gram-positive agents in cases of catheter-related infections or cellulitis [2]. Beyond the lack of a clear survival benefit, the routine use of vancomycin is discouraged due to the risk of emerging resistance and potential side effects, including vancomycin-associated neutropenia and kidney injury [99].

3.2. Effects on microbiome and GVHD

Vancomycin exhibits antimicrobial activity against both aerobic and anaerobic gram-positive bacteria. While oral vancomycin significantly alters the gut microbiome [32], intravenous (IV) vancomycin appears to have minimal impact. A German study found that IV vancomycin did not reduce urinary 3-indoxyl sulfate, a biomarker of microbiome diversity [39]. Similarly, microbiome analyses following IV vancomycin

administration demonstrated minimal effects on gut microbiome composition [32]. This limited impact on the microbiome may be due to vancomycin's pharmacokinetic properties. Orally administered vancomycin has minimal systemic absorption, while IV vancomycin is primarily eliminated renally, with only a very small fraction excreted via bile [100]. However, vancomycin has been detected in stool samples of hospitalized patients with diarrhea, and small amounts have been measurable in bile [101,102]. There are limited reports examining the relationship between IV vancomycin use and GVHD. A few studies have suggested an association between glycopeptide use before bone marrow transplantation (BMT) and an increased risk of GVHD [37,42].

3.3. Resistance and infection

The emergence of vancomycin-resistant Enterococci (VRE) is one of the most significant infectious threats in allogeneic and autologous HSCT recipients. Antibiotic use, including not only vancomycin but also fluoroquinolones, third-generation cephalosporins, anaerobic-targeting antibiotics, and aminoglycosides, has been linked to the emergence of VRE [103–105]. A recent study by Nørgaard investigated the impact of various antibiotic treatments on the gut microbiome and resistome in HSCT recipients using a large number of fecal samples. IV vancomycin exposure was associated with increased resistance to aminoglycosides, trimethoprim, glycopeptides, and multidrug efflux pump mechanisms [33]. The Enterococcal *vanA* gene, which confers high-level vancomycin resistance, was linked to both oral and IV vancomycin use but was more strongly associated with piperacillin-tazobactam and meropenem exposure [33]. For VRE colonized patients, VRE-active agent such as linezolid are often used empirically. However, in a recent study, there was no significant difference in incidence of VRE bacteremia between VRE colonized patients who received versus did not receive empiric therapy, primarily with linezolid [106].

3.4. Practice point – antimicrobial stewardship and de-escalation

Given the limited evidence supporting the routine use of additional gram-positive agents, their use should be restricted to patients who meet clear clinical criteria, such as those with cellulitis or catheter-related infections. Once initiated, gram-positive coverage should be closely monitored and reevaluated to ensure appropriate de-escalation. The IDSA recommends discontinuing additional gram-positive agents within 2–3 days if no susceptible gram-positive bacteria are identified [98].

4. Fluoroquinolones and other antibiotic prophylaxis

4.1. The role of prophylaxis

Compared to the empiric treatment of FN with broad-spectrum antibiotics, the evidence supporting prophylactic antibiotic use is less robust. Although prophylaxis has been

the standard of care in many transplant centers worldwide, the role of antibiotic prophylaxis remains controversial. Early approaches focused on gastrointestinal decontamination using non-absorbable antibiotics. These studies, conducted primarily in patients with acute leukemia undergoing intensive chemotherapy, failed to demonstrate clear benefits in terms of infection prophylaxis [107,108].

Subsequently, fluoroquinolone prophylaxis was extensively studied as a strategy for infection prevention. A 2012 meta-analysis demonstrated a mortality benefit with fluoroquinolone prophylaxis compared to placebo or no treatment, reporting significant reductions in both all-cause mortality (RR 0.54) and infection-related mortality (RR 0.51) [109]. These benefits were observed only with levofloxacin and ciprofloxacin but not with norfloxacin. However, this meta-analysis included individuals with any cancer, with only 33 out of 109 trials involved allogeneic or autologous HSCT recipients. Nonetheless, the findings influenced many international guidelines that currently recommend fluoroquinolone prophylaxis, including those from the United States (ASCO/ISDA) [98,110,111] and Germany [112].

Subsequent meta-analyses confirmed reduced bacteremia and FN but have not shown a survival benefit [113–116]. The lack of survival benefit may be attributed to higher rates of fluoroquinolone resistance and lower mortality in recent studies compared to earlier trials [117]. Notably, a meta-analysis focused exclusively on HSCT recipients showed that prophylaxis reduced infections and fever but did not improve survival [114]. Several studies have been conducted after these meta-analyses. In a 2018 randomized controlled trial, the effect of levofloxacin prophylaxis on bacteremia was assessed in pediatric patients with acute leukemia or undergoing HSCT [118]. While levofloxacin significantly reduced bacteremia in patients with acute leukemia, its effect was not significant in HSCT recipients. Similarly, a large multicenter surveillance study including 8,755 patients found that fluoroquinolone prophylaxis reduced gram-negative bacteremia and improved survival in patients with acute leukemia but had unclear benefits in HSCT recipients [119]. A retrospective study in children ($N=443$) comparing pre- and post- introduction of fluoroquinolone prophylaxis found reduced bloodstream infections, CDIs, and acute GVHD [120]. An adult retrospective study ($N=254$) comparing pre- and post- discontinuation of fluoroquinolone prophylaxis found that fluoroquinolone prophylaxis (primarily ciprofloxacin) reduced gram-negative bloodstream infections (21% vs. 33%, $p=0.027$) but did not impact overall mortality [121]. Similar results were demonstrated in other recent studies [56,122].

Overall, fluoroquinolone prophylaxis reduces gram-negative infections but shows no survival benefit in patients undergoing HSCT. Moreover, it may increase gram-positive infections [119,123]. Prophylaxis is also ineffective in patients colonized with resistant organisms. Satlin *et al.* reported that 32% of HSCT recipients colonized with ESBL-producing *Enterobacteriaceae* developed ESBL-E bacteremia during fluoroquinolone prophylaxis [124]. Similarly, nearly one-third of fluoroquinolone-resistant Enterobacterales carriers developed gram-negative bacteremia despite levofloxacin prophylaxis [125]. Due to the uncertainty surrounding benefits and risks

of fluoroquinolone prophylaxis, international guidelines vary in their recommendations. While the guidelines from the U.S. (ASCO/ISDA) [98,110,111] and Germany [112] support its use, Australian [126] and European Society for Medical Oncology (ESMO) guidelines [2] advise against the routine use of fluoroquinolone prophylaxis. In practice, fluoroquinolone prophylaxis is widely implemented – 57% of European centers and 85% of U.S. centers reporting its use [82,83].

4.2. Effects of fluoroquinolone prophylaxis on the microbiome and GVHD

In pre-clinical models, gastrointestinal decontamination was studied as strategy to potentially prevent GVHD, and these studies demonstrated that GVHD can be prevented by restriction or elimination of the microbiome [127,128]. Vossen *et al.* reviewed 65 children receiving allo-HSCT who underwent complete gastrointestinal decontamination with high-dose non-absorbable antimicrobials and compared them to those receiving selective gastrointestinal decontamination using either absorbable antimicrobials or low-dose non-absorbable antimicrobials [129]. They found that complete decontamination was superior in preventing GVHD and infectious complications. In a subsequent analysis of 112 pediatric patients, the same group compared the incidence of acute GVHD between those who achieved successful gastrointestinal decontamination and those who did not [130]. GVHD occurred in only 1 out of 57 patients in the successful decontamination group, compared to 8 out of 55 in the unsuccessful group ($p=0.013$). However, the study did not demonstrate a survival benefit, possibly due to the extremely low incidence of GVHD in both groups.

Beelen *et al.* retrospectively analyzed 194 predominantly adult allo-HSCT recipients treated under strict protective isolation with intestinal antimicrobial decontamination. They concluded that sustained suppression of intestinal anaerobic bacteria helps prevent acute GVHD [131]. In a prospective randomized controlled trial (RCT), Beelen *et al.* compared metronidazole plus ciprofloxacin versus ciprofloxacin alone and found that the combination was more effective in preventing GVHD [132]. It is not clear why suppression of anaerobes helps reducing GVHD in these studies, while broad-spectrum antibiotics with anaerobe cover was associated with increased GVHD [30,31]. It may relate to inflammation driven by aberrant host response to anaerobic bacteria also seen in inflammatory bowel disease [133].

More recently, Weber *et al.* retrospectively compared rifaximin with a combination of ciprofloxacin and metronidazole [39,46]. Rifaximin is a rifamycin derivative with broad-spectrum activity and negligible absorption. Their findings suggested that rifaximin helps preserve gut microbiome diversity, as indicated by higher concentrations of 3-indoxyl sulfate, a marker of microbial diversity. Patients receiving rifaximin also exhibited lower transplant-related mortality and improved overall survival. However, a small prospective study ($N=38$) did not replicate these findings [134].

Either ciprofloxacin or levofloxacin, with minimal anaerobic cover, is commonly used as fluoroquinolone prophylaxis. These fluoroquinolones appear to have a more limited impact

on gut microbiome. Morjaria *et al.* and Daoud-Asfour *et al.* both reported that ciprofloxacin had minimal effects on microbiome, while Rashid *et al.* reported decreases in certain bacteria in healthy volunteers [32,135,136]. With levofloxacin, increased microbiome diversity and increased *Blautia*, beneficial bacteria, were reported [137,138]. Clinically, several reports indicate that fluoroquinolone prophylaxis does not increase GVHD incidence [36,120,135]. Stoma *et al.* analyzed detailed effects of fluoroquinolone prophylaxis on gut colonization and showed significant decreases of abundance in gram-negative pathogens such as *Klebsiella spp.*, *Citrobacter spp.* and *Enterobacter spp.*, but not *Escherichia coli*, suggesting that fluoroquinolones may decrease gram negative bacteremia by changing gastrointestinal colonization [56].

4.3. Fluoroquinolones and antibiotic resistance, loss of colonisation resistance, and CDI

Fluoroquinolone prophylaxis is associated with increased rates of antibiotic resistance [33]. In hematologic and HSCT recipients, fluoroquinolone prophylaxis has been also associated with meropenem-resistant *P. aeruginosa*, likely due to mutations that enhance efflux pump activity [139]. A recent study by Clerici *et al.* showed that discontinuation of fluoroquinolone prophylaxis was associated with significantly higher susceptibility to piperacillin-tazobactam, carbapenems and fluoroquinolones [122]. In addition, Leardini *et al.* compared clinical outcomes and antibiotic resistance between pediatric allo-HSCT recipients who received levofloxacin prophylaxis and those who did not [140]. While there were no significant differences in HSCT outcomes, including infection rates, patients who received prophylaxis had a significantly higher incidence of quinolone-resistant bacterial strains ($p=0.001$). Similarly, Stern *et al.* reported that gram-negative bacteremia was more frequently resistant to fluoroquinolones in adult patients receiving fluoroquinolone prophylaxis compared to those who did not (68.9% vs. 41.6%, $p=0.021$) [121].

A recent U.S. study found that only 49% of gram-negative isolates remained susceptible to fluoroquinolones in patients with hematological malignancies, 57% of whom had received fluoroquinolone prophylaxis [141]. Additionally, a large international study spanning 25 countries revealed alarmingly high gram-negative resistance rates among patients undergoing allo-HSCT (62.9% in adults), with fluoroquinolone resistance exceeding 50% in *Escherichia coli* and *Klebsiella pneumoniae* [142]. The rate of fluoroquinolone-resistant gram-negative bacilli were significantly higher in centers that routinely used fluoroquinolone prophylaxis (79% vs. 50% in allo-HSCT, 74% vs. 24% in autologous HSCT). A systematic review also identified an increased prevalence of fluoroquinolone-resistant *Enterobacteriales* in bacteremia among patients who received fluoroquinolone prophylaxis [113]. However, other meta-analyses did not consistently find significant resistance development following fluoroquinolone prophylaxis [109,114–116].

Fluoroquinolone prophylaxis has also been linked to an increased risk of CDI in the general patient population [143]. However, its impact on allogeneic and autologous HSCT recipients remains unclear due to confounding factors such as chemotherapy and concurrent antibiotic use. Studies in HSCT (both

allogeneic and autologous) recipients do not consistently show an increase in CDI following fluoroquinolone prophylaxis [118,120,144,145]. Fluoroquinolones reduce bacteremia and FN [113–116], which generally leads to less exposure to broad-spectrum antibiotics. This may contribute to no increase or even a decrease in CDI in some studies. Due to established data of increased CDI in other populations, caution is recommended in HSCT until more comprehensive studies are performed.

4.4. Review of fluoroquinolone prophylaxis

Current guidelines recommending fluoroquinolone prophylaxis, including those from the U.S. and Germany [98,110–112], are primarily based on a 2012 meta-analysis [109]. However, when included studies were conducted, fluoroquinolone resistance was less prevalent, and the role of the microbiome in HSCT outcomes was not as well understood. While fluoroquinolone prophylaxis may provide short-term benefits in reducing infection and FN, it may also contribute to long-term harm through the development of AMR and microbiome disruption. Therefore, its routine use should be carefully reconsidered, particularly in patients undergoing HSCT in hospitals where rapid access to empiric antibiotic therapy is available [126].

Several important questions remain unanswered, i.e.: Does fluoroquinolone prophylaxis improve HSCT outcomes by reducing the need for broad-spectrum antibiotic exposure? What is the role of fluoroquinolone prophylaxis in allo-HSCT patients receiving post-transplant cyclophosphamide (PTCy)? In this setting, neutropenia duration and infection risk appear to be higher than in conventional allo-HSCT. What is the role of rifaximin as an alternative to fluoroquinolone for prevention of FN?

5. Conclusions

There is increasing awareness of the impact of antibiotics on the intestinal microbiome. In allo-HSCT, depletion of anaerobic commensal microbiota has been linked to an increased risk of GVHD and mortality. Large retrospective observational studies have suggested that broad-spectrum antibiotics with anaerobic coverage, such as piperacillin-tazobactam and carbapenems, are associated with poorer clinical outcomes compared to cefepime, which has minimal anaerobic activity. However, these studies have not fully characterized infectious outcomes, particularly infections caused by anaerobic bacteria. Therefore, there is a need for large prospective randomized controlled trials (RCTs) to evaluate both GVHD and infection outcomes, with close microbiome monitoring.

The impact of fluoroquinolone prophylaxis and additional gram-positive intravenous antibiotics on the microbiome, GVHD and survival remains unclear. However, the overuse of these antibiotics is associated with the development of AMR. Despite the fact that most guidelines do not recommend the routine use of additional gram-positive agents, and that guidelines are not consistent regarding the use of fluoroquinolone prophylaxis, many centers continue to use this strategy. Regular updates to guidelines, improved adherence, and antimicrobial

stewardship (AMS) strategies such as de-escalation and shorter antibiotic durations are necessary to mitigate these risks.

6. Expert opinion

Beyond well-established concerns such as AMR and CDI, increasing evidence highlights the impact of antibiotic-associated dysbiosis on post-HSCT outcomes, particularly allo-HSCT. The key strategy to mitigate these risks is personalized, risk-adaptive AMS (Figure 1). This includes selective use and de-escalation of antibiotics and microbiome preservative antibiotic choice. Both require integration into routine HSCT care and further validation through clinical research.

6.1. Risk-adaptive antimicrobial stewardship

Although HSCT is a highly protocol-driven treatment, AMS in HSCT should shift from a uniform approach to one that incorporates personalized risk profiles, local resistance patterns, and microbiome preservation. Personalized risk assessment includes routine surveillance of colonization by AMR organisms (VRE, CPE, ESBL and fluoroquinolone-resistant bacteria), and other risk factors such as age, past infections and comorbidities. Pre-HSCT microbiome data could be included in future clinical practice to guide HSCT management.

Additional gram-positive coverage should be limited to patients who meet clinical criteria and discontinued promptly based on microbiological data and clinical response. Frequent reassessment and de-escalation are essential. Close monitoring of renal function and drug exposure is important, to minimize the risk of renal impairment. Therapeutic drug monitoring, especially AUC based monitoring is useful if vancomycin is continued. Fluoroquinolone prophylaxis should be used selectively. Although HSCT recipients are generally high-risk, subsets, such as inpatients with timely access to empiric antibiotics or those receiving non-myeloablative conditioning, may benefit from individualized risk assessment. High risk patients, who are at risk of developing complicated FN may benefit from fluoroquinolone prophylaxis. These high-risk patients include elderly, those with significant co-morbidities and patients with a history of serious bacterial infections. Rising fluoroquinolone resistance and improved sepsis management call for frequent updates to prophylaxis guidelines. If fluoroquinolone prophylaxis is used, timing matters – initiating at the onset of neutropenia (rather than commencement of chemotherapy) may limit unnecessary exposure.

Empirical broad-spectrum antibiotics for FN remain standard, but judicious use and early de-escalation are more controversial. Evidence supports safe early discontinuation in afebrile, clinically stable patients [45,84,85]. Therefore, careful assessment of an individual patient is essential. Importantly, broad-spectrum antibiotics should not be used routinely in non-neutropenic fever, but instead each patient evaluated for focal infections. Routine AMR colonization screening prior to transplant may enable more targeted antibiotic use. Antibiotic selection should consider both efficacy and microbiome impact. Cefepime and piperacillin-tazobactam remain first-line options in FN. Cefepime offers better CNS penetration and coverage of AmpC producers, while piperacillin-

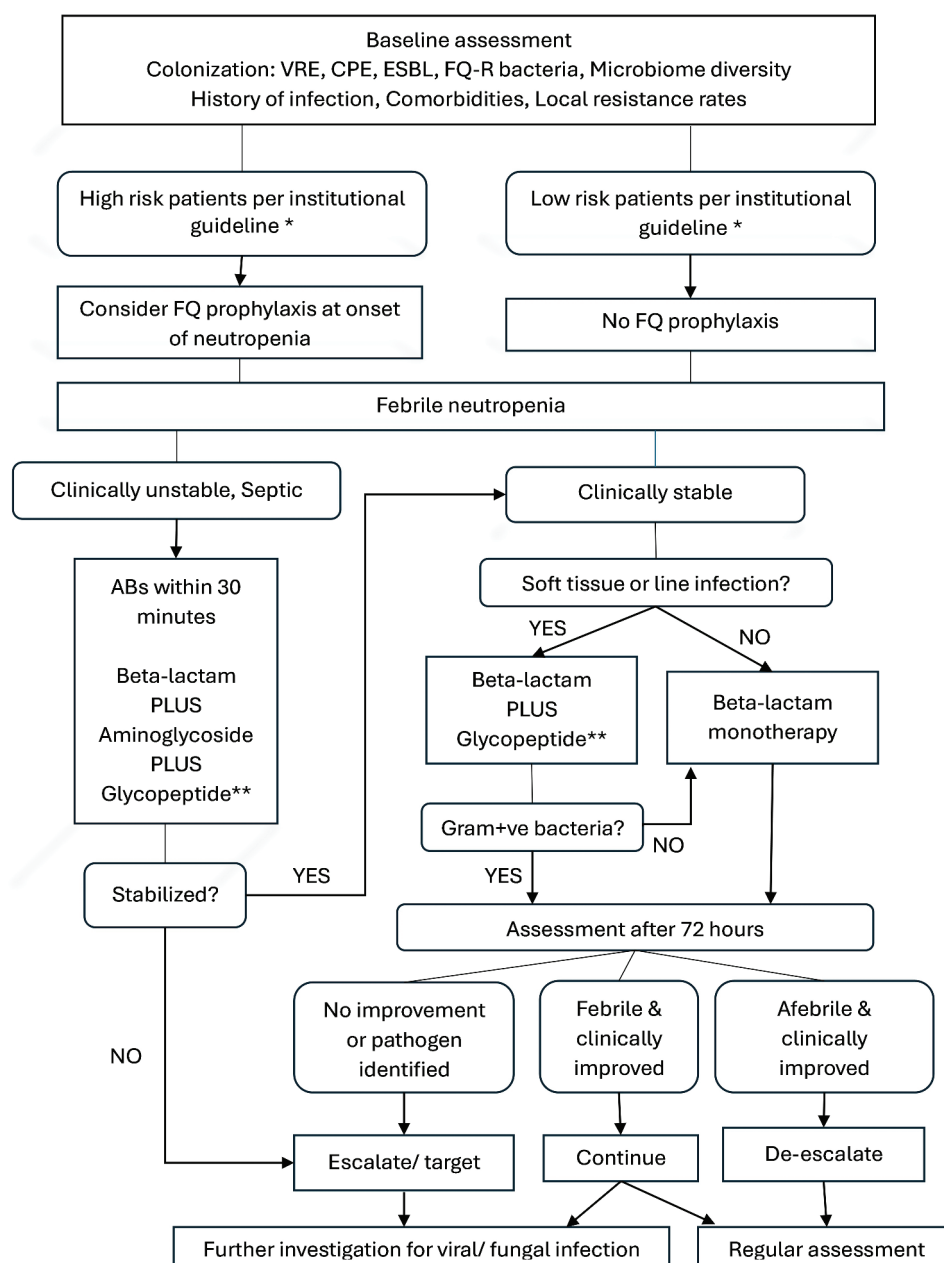


Figure 1. Personalized AMS for prevention and treatment of FN. * risk factors include age, comorbidities, past infections and outpatient transplant. Consider institutional resistance pattern and patient's colonization. ** consider VRE colonization.

tazobactam provides anaerobic and Enterococcal cover. Therefore, the choice should be based on the clinical features, including the risk of anaerobic infection.

6.2. Future directions and considerations

Given that antibiotic use is often unavoidable in HSCT, strategies that protect and restore gut microbiome integrity are especially valuable. Although microbiome-targeted interventions outside of the scope of this review, nonspecific microbiome modulation strategies, such as dietary changes, prebiotics, probiotics and fecal microbiota transplantation, are increasingly being explored [20–25]. Microbiome-targeted interventions are still investigational. There have not been large scale clinical trials and how to best protect or restore healthy microbiome is still unclear. High

quality clinical trials are required to identify the effective interventions. Pre-transplant microbiome composition may predict transplant outcomes, suggesting that early intervention could be beneficial. Preservation and restoration of healthy gut microbiome not only reduces GVHD, but also likely to reduce AMR and CDI. Other emerging experimental approaches include gut-targeted antibiotic deactivation using oral beta-lactamases or activated charcoal. However, luminal beta-lactamases are highly specific and limited in scope, while activated charcoal may have overly broad, nonselective effects on the gut environment and absorption of other drugs. Nevertheless, these new approaches may be used in conjunction with personalized AMS strategy in the future. Future antibiotic development should address not only antimicrobial activity and resistance but also microbiome preservation and gut pharmacokinetics.

Other issues include access to next-generation sequencing for microbiome characterization. Whilst it is widely available in research setting in the developed world, there are few validated clinical tests. Most major retrospective and prospective studies have been conducted in a small number of leading HSCT centers, mainly in the U.S. and Germany. However, AMR patterns, microbiome profiles, prophylaxis strategies, and empiric treatment regimens vary across regions. Therefore, more studies from diverse geographical settings are needed to make existing evidence more generalizable. In addition, allo-HSCT practices are undergoing significant changes globally, particularly with the increasing use of post-transplant cyclophosphamide (PTCy). Large cohort studies evaluating broad-spectrum antibiotic effects on GVHD have focused on conventional allo-HSCT and have generally excluded T-cell-depleted transplants [30,31]. Since PTCy is associated with reduced GVHD but prolonged neutropenia, the impact of broad-spectrum antibiotics in this setting may differ from earlier findings. Similarly, the efficacy and risks of fluoroquinolone prophylaxis may need to be reevaluated in the PTCy era.

In summary, advancing patient outcomes in allo-HSCT will require maintenance of a healthy gut microbiome and personalized, risk-adapted AMS strategies. In addition, future research should provide additional strategies to protect and restore the gut microbiome. These strategies will help strike a better balance between infection management and microbiome preservation in allo-HSCT recipients.

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References

Papers of special note have been highlighted as either of interest (*) or of considerable interest () to readers.**

1. Celebi H, Akan H, Akçağlayan E, et al. Febrile neutropenia in allogeneic and autologous peripheral blood stem cell

- transplantation and conventional chemotherapy for malignancies. *Bone Marrow Transplant.* 2000;26(2):211–214. doi: [10.1038/sj.bmt.1702503](#)
2. Klastersky J, De Naurois J, Rolston K, et al. Management of febrile neutropenia: ESMO clinical practice guidelines. *Ann Of Oncol.* 2016;27:v111–v118. doi: [10.1093/annonc/mdw325](#)
3. Daniels LM, Durani U, Barreto JN, et al. Impact of time to antibiotic on hospital stay, intensive care unit admission, and mortality in febrile neutropenia. *Supportive Care In Cancer.* 2019;27(11):4171–4177. doi: [10.1007/s00520-019-04701-8](#)
4. Naghavi M, Vollset SE, Ikuta KS, et al. Global burden bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050. *Lancet.* 2024;404(10459):1199–1226. doi: [10.1016/S0140-6736\(24\)01867-1](#)
5. van Werkhoven Ch, Ducher A, Berkell M, et al. Incidence and predictive biomarkers of clostridioides difficile infection in hospitalized patients receiving broad-spectrum antibiotics. *Nat Commun.* 2021;12(1):2240. doi: [10.1038/s41467-021-22269-y](#)
6. Zhang Z, Wang J, Wang H, et al. Association of infant antibiotic exposure and risk of childhood asthma: a meta-analysis. *World Allergy Organ J.* 2021;14(11):100607. doi: [10.1016/j.waojou.2021.100607](#)
7. Crespin A, Le Bescop C, de Gunzburg J, et al. A systematic review and meta-analysis evaluating the impact of antibiotic use on the clinical outcomes of cancer patients treated with immune checkpoint inhibitors. *Front Oncol.* 2023;13:1075593. doi: [10.3389/fonc.2023.1075593](#)
8. Derosa L, Routy B, Desilets A, et al. Microbiota-centered interventions: the next breakthrough in immuno-Oncology? *Cancer Discov.* 2021;11(10):2396–2412. doi: [10.1158/2159-8290.CD-21-0236](#)
9. Akagbosu B, Tayyebi Z, Shibu G, et al. Novel antigen-presenting cell imparts treg-dependent tolerance to gut microbiota. *Nature.* 2022;610(7933):752–760. doi: [10.1038/s41586-022-05309-5](#)
10. Flint HJ, Scott KP, Duncan SH, et al. Microbial degradation of complex carbohydrates in the gut. *Gut Microbes.* 2012;3(4):289–306. doi: [10.4161/gmic.19897](#)
11. LeBlanc JG, Milani C, De Giori GS, et al. Bacteria as vitamin suppliers to their host: a gut microbiota perspective. *Curr Opin Biotechnol.* 2013;24(2):160–168. doi: [10.1016/j.copbio.2012.08.005](#)
12. Mathewson ND, Jenq R, Mathew AV, et al. Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease. *Nat Immunol.* 2016;17(5):505–513. doi: [10.1038/ni.3400](#)
13. Belkaid Y, Hand Timothy W. Role of the microbiota in immunity and inflammation. *Cell.* 2014;157(1):121–141. doi: [10.1016/j.cell.2014.03.011](#)
14. Pickard JM, Zeng MY, Caruso R, et al. Gut microbiota: role in pathogen colonization, immune responses, and inflammatory disease. *Immunol Rev.* 2017;279(1):70–89. doi: [10.1111/imr.12567](#)
15. Cryan JF, O'Riordan KJ, Cowan CS, et al. The microbiota-gut-brain axis. *Physiol Rev.* 2019;99(4):1877–2013. doi: [10.1152/physrev.00018.2018](#)
16. Jenq RR, Ubeda C, Taur Y, et al. Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J Exp Med.* 2012;209(5):903–911. doi: [10.1084/jem.20112408](#)
17. Taur Y, Jenq RR, Perales M-A, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood J Am Soc Hematol.* 2014;124(7):1174–1182. doi: [10.1182/blood-2014-02-554725](#)
- One of the first studies that identified microbiome – mortality association.**
18. Weber D, Oefner PJ, Hiergeist A, et al. Low urinary indoxyl sulfate levels early after transplantation reflect a disrupted microbiome and are associated with poor outcome. *Blood, J Am Soc Hematol.* 2015;126(14):1723–1728. doi: [10.1182/blood-2015-04-638858](#)
19. Peled JU, Gomes AL, Devlin SM, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med.* 2020;382(9):822–834. doi: [10.1056/NEJMoa1900623](#)
- A global study showing microbiome can predict HSCT outcomes.**

20. Shono Y, van den Brink MRM, van den Brink MRM. Gut microbiota injury in allogeneic haematopoietic stem cell transplantation. *Nat Rev Cancer*. 2018;18(5):283–295. doi: [10.1038/nrc.2018.10](#)
21. Andermann TM, Rezvani A, Bhatt AS. Microbiota manipulation with prebiotics and probiotics in patients undergoing stem cell transplantation. *Curr Hematol Malig Rep*. 2016;11(1):19–28. doi: [10.1007/s11899-016-0302-9](#)
22. Malard F, Gaugler B, Mohty M. Faecal microbiota transplantation in patients with haematological malignancies undergoing cellular therapies: from translational research to routine clinical practice. *Lancet Haematol*. 2022;9(10):e776–e785. doi: [10.1016/S2352-3026\(22\)00223-X](#)
23. Muratore E, Leardini D, Baccelli F, et al. Nutritional modulation of the gut microbiome in allogeneic hematopoietic stem cell transplantation recipients. *Front Nutr*. 2022;9:993668. doi: [10.3389/fnut.2022.993668](#)
24. Sorbara MT, Pamer EG. Microbiome-based therapeutics. *Nat Rev Microbiol*. 2022;20(6):365–380. doi: [10.1038/s41579-021-00667-9](#)
25. van Lier Yf, Vos J, Blom B, et al. Allogeneic hematopoietic cell transplantation, the microbiome, and graft-versus-host disease. *Gut Microbes*. 2023;15(1):2178805. doi: [10.1080/19490976.2023.2178805](#)
26. Chumbita M, Puerta-Alcalde P, Gudiol C, et al. Impact of empirical antibiotic regimens on mortality in neutropenic patients with bloodstream infection presenting with septic shock. *Antimicrob Agents Chemother*. 2022;66(2):e0174421. doi: [10.1128/AAC.01744-21](#)
27. Bow EJ, Rotstein C, Noskin GA, et al. A randomized, open-label, multicenter comparative study of the efficacy and safety of piperacillin-tazobactam and cefepime for the empirical treatment of febrile neutropenic episodes in patients with hematologic malignancies. *Clin Infect Dis*. 2006;43(4):447–459. doi: [10.1086/505393](#)
28. Raad EC II, Hachem RY, Hachem RY, et al. Treatment of febrile neutropenic patients with cancer who require hospitalization: a prospective randomized study comparing imipenem and cefepime. *Cancer*. 2003;98(5):1039–1047. doi: [10.1002/cncr.11613](#)
29. Uygun V, Karasu GT, Ogunc D, et al. Piperacillin/Tazobactam versus cefepime for the empirical treatment of pediatric cancer patients with neutropenia and fever: a randomized and open-label study. *Pediatr Blood Cancer*. 2009;53(4):610–614. doi: [10.1002/pbc.22100](#)
30. Shono Y, Docampo MD, Peled JU, et al. Increased GVHD-related mortality with broad-spectrum antibiotic use after allogeneic hematopoietic stem cell transplantation in human patients and mice. *Sci Transl Med*. 2016;8(339):ra33971–ra33971. doi: [10.1126/scitranslmed.aaf2311](#)
- **A key study discovered that antibiotics with anaerobe cover is associated with worse HSCT outcomes.**
31. Rashidi A, Gao F, Fredricks DN, et al. Analysis of antibiotic exposure and development of acute graft-vs-host disease following allogeneic hematopoietic cell transplantation. *JAMA Network Open*. 2023;6(6):e2317188–e2317188. doi: [10.1001/jamanetworkopen.2023.17188](#)
32. Morjaria S, Schluter J, Taylor BP, et al. Antibiotic-induced shifts in fecal microbiota density and composition during hematopoietic stem cell transplantation. *Infect Immun*. 2019;87(9). doi: [10.1128/IAI.00206-19](#)
33. Nørgaard JC, Jørgensen M, Moestrup KS, et al. Impact of antibiotic treatment on the gut microbiome and its resistome in hematopoietic stem cell transplant recipients. *J Infect Dis*. 2023;228(1):28–36. doi: [10.1093/infdis/jiad033](#)
34. Elgarten CW, Li Y, Getz KD, et al. Broad-spectrum antibiotics and risk of graft-versus-host disease in pediatric patients undergoing transplantation for acute leukemia: association of carbapenem use with the risk of acute graft-versus-host disease. *Transpl Cell Ther*. 2021;27(2):e177.1–e177.8. doi: [10.1016/j.jct.2020.10.012](#)
35. Gavrilaki M, Sakellari I, Anagnostopoulos A, et al. The impact of antibiotic-mediated modification of the intestinal microbiome on outcomes of allogeneic hematopoietic cell transplantation: systematic review and meta-analysis. *Biol Blood Marrow Transplant*. 2020;26(9):1738–1746. doi: [10.1016/j.bbmt.2020.05.011](#)
36. Hanks CR, Slain D, Kanate AS, et al. Impact of anti-anaerobic antibiotic activity on graft-versus-host disease in allogeneic hematopoietic stem cell transplant (HSCT) recipients at an institution that utilizes antibiotic cycling. *Transpl Infect Dis*. 2021;23(4):e13676. doi: [10.1111/tid.13676](#)
37. Hidaka D, Hayase E, Shiratori S, et al. The association between the incidence of intestinal graft-vs-host disease and antibiotic use after allogeneic hematopoietic stem cell transplantation. *Clin Transplant*. 2018;32(9):e13361. doi: [10.1111/ctr.13361](#)
38. Lee SE, Lim JY, Ryu DB, et al. Alteration of the intestinal microbiota by broad-spectrum antibiotic use correlates with the occurrence of intestinal graft-versus-host disease. *Biol Blood Marrow Transplant*. 2019;25(10):1933–1943. doi: [10.1016/j.bbmt.2019.06.001](#)
39. Weber D, Hiergeist A, Weber M, et al. Detrimental effect of broad-spectrum antibiotics on intestinal microbiome diversity in patients after allogeneic stem cell transplantation: lack of commensal sparing antibiotics [article]. *Clin Infect Dis*. 2019;68(8):1303–1310. doi: [10.1093/cid/ciy711](#)
40. Weber D, Jenq RR, Peled JU, et al. Microbiota disruption induced by early use of broad-spectrum antibiotics is an Independent risk factor of outcome after allogeneic stem cell transplantation [article]. *Biol Blood Marrow Transplant*. 2017;23(5):845–852. doi: [10.1016/j.bbmt.2017.02.006](#)
- **A German study demonstrating the timing of antibiotic use is important.**
41. Holler E, Butzhammer P, Schmid K, et al. Metagenomic analysis of the stool microbiome in patients receiving allogeneic stem cell transplantation: loss of diversity is associated with use of systemic antibiotics and more pronounced in gastrointestinal graft-versus-host disease. *Biol Blood Marrow Transplant*. 2014;20(5):640–645. doi: [10.1016/j.bbmt.2014.01.030](#)
42. Lee MW, Yeon SH, Heo BY, et al. Impact of pre-transplant use of antibiotics on the graft-versus-host disease in adult patients with hematological malignancies. *Hematology*. 2021;26(1):96–102. doi: [10.1080/16078454.2021.1872957](#)
43. Meedt E, Hiergeist A, Gessner A, et al. Prolonged suppression of butyrate-producing bacteria is associated with acute gastrointestinal graft-vs-host disease and transplantation-related mortality after allogeneic stem cell transplantation [article]. *Clin Infect Dis*. 2022;74(4):614–621. doi: [10.1093/cid/ciab500](#)
44. Nishi K, Kanda J, Hishizawa M, et al. Impact of the use and type of antibiotics on acute graft-versus-host disease. *Biol Blood Marrow Transplant*. 2018;24(11):2178–2183. doi: [10.1016/j.bbmt.2018.06.031](#)
45. Weber D, Hiergeist A, Weber M, et al. Restrictive versus permissive use of broad-spectrum antibiotics in patients receiving allogeneic stem cell transplantation and with early fever due to Cytokine release syndrome: evidence for beneficial microbiota protection without increase in infectious complications [article]. *Clin Infect Dis*. 2023;77(10):1432–1439. doi: [10.1093/cid/ciad389](#)
46. Weber D, Oefner PJ, Dettmer K, et al. Rifaximin preserves intestinal microbiota balance in patients undergoing allogeneic stem cell transplantation [article]. *Bone Marrow Transplant*. 2016;51(8):1087–1092. doi: [10.1038/bmt.2016.66](#)
47. Devaux CA, Million M, Raoult D. The Butyrogenic and lactic bacteria of the gut microbiota determine the outcome of allogeneic hematopoietic cell transplant. *Front Microbiol*. 2020;11:1642. doi: [10.3389/fmicb.2020.01642](#)
48. Nalle SC, Zuo L, Ong MLDM, et al. Graft-versus-host disease propagation depends on increased intestinal epithelial tight junction permeability. *J Clin Invest*. 2019;129(2):902–914. doi: [10.1172/JCI98554](#)
49. Payen M, Nicolis I, Robin M, et al. Functional and phylogenetic alterations in gut microbiome are linked to graft-versus-host disease severity. *Blood Adv*. 2020;4(9):1824–1832. doi: [10.1182/bloodadvances.2020001531](#)
50. Koyama M, Mukhopadhyay P, Schuster IS, et al. MHC class II antigen presentation by the intestinal epithelium initiates graft-versus-host disease and is influenced by the microbiota. *Immunity*. 2019;51(5):885–898.e7. doi: [10.1016/j.immuni.2019.08.011](#)

51. Koyama M, Hippe DS, Srinivasan S, et al. Intestinal microbiota controls graft-versus-host disease independent of donor-host genetic disparity [article]. *Immunity*. 2023;56(8):1876–1893.e8. doi: 10.1016/j.immuni.2023.06.024
52. Hayase E, Hayase T, Jamal MA, et al. Mucus-degrading bacteroides link carbapenems to aggravated graft-versus-host disease. *Cell*. 2022;185(20):3705–3719.e14. doi: 10.1016/j.cell.2022.09.007
53. Hayase E, Hayase T, Mukherjee A, et al. *Bacteroides ovatus* alleviates dysbiotic microbiota-induced graft-versus-host disease. *Cell Host Microbe*. 2024;32(9):1621–1636.e6. doi: 10.1016/j.chom.2024.08.004
54. Merrick B, Sergaki C, Edwards L, et al. Modulation of the gut microbiota to control antimicrobial resistance (AMR)-A narrative review with a focus on faecal microbiota transplantation (FMT). *Infect Dis Rep*. 2023;15(3):238–254. doi: 10.3390/idr15030025
55. Carlet J. The gut is the epicentre of antibiotic resistance. *Antimicrob Resist Infect Control*. 2012;1(1):39. doi: 10.1186/2047-2994-1-39
56. Stoma I, Littmann ER, Peled JU, et al. Compositional flux within the intestinal microbiota and risk for bloodstream infection with gram-negative bacteria. *Clin Infect Dis*. 2020;73(11):e4627–e4635. doi: 10.1093/cid/ciaa068.
- **A large study explored gram negative colonisation and bacteraemia, with effects of fluoroquinolone prophylaxis.**
57. Kelly MS, Ward DV, Severyn CJ, et al. Gut colonization preceding mucosal barrier injury bloodstream infection in pediatric hematopoietic stem cell transplantation recipients. *Biol Blood Marrow Transplant*. 2019;25(11):2274–2280. doi: 10.1016/j.bbmt.2019.07.019
58. Anthony WE, Burnham CD, Dantas G, et al. The gut microbiome as a reservoir for antimicrobial resistance. *J Infect Dis*. 2021;223(12 Suppl 2):S209–S213. doi: 10.1093/infdis/jiaa497
59. Bilinski J, Robak K, Peric Z, et al. Impact of gut colonization by antibiotic-resistant bacteria on the outcomes of allogeneic hematopoietic stem cell transplantation: a retrospective, single-center study. *Biol Blood Marrow Transplant*. 2016;22(6):1087–1093. doi: 10.1016/j.bbmt.2016.02.009
60. Pilmsis B, Le Monnier A, Gut Microbiota ZJ. Antibiotic therapy and antimicrobial resistance: a narrative review. *Microorganisms*. 2020;8(2):269. doi: 10.3390/microorganisms8020269
61. Matzaras R, Nikopoulou A, Protonotariou E, et al. Gut microbiota modulation and prevention of Dysbiosis as an alternative approach to antimicrobial resistance: a narrative review. *Yale J Biol Med*. 2022;95(4):479–494. doi: 10.1101/gr.096651.109
62. Kaundal S, Jandial A, Singh H, et al. Impact of broad-spectrum antibiotic exposures and multidrug-resistant gram-negative bacteraemia on hematopoietic cell transplantation outcomes. *Transpl Infect Dis*. 2021;23(5):e13717. doi: 10.1111/tid.13717
63. Gudiol C, Albasanz-Puig A, Laporte-Amargós J, et al. Clinical predictive Model of multidrug resistance in neutropenic cancer patients with bloodstream infection due to *Pseudomonas aeruginosa*. *Antimicrob Agents Chemother*. 2020;64(4):10–19. doi: 10.1128/AAC.02494-19
64. Heston SM, Young RR, Jenkins K, et al. The effects of antibiotic exposures on the gut resistome during hematopoietic cell transplantation in children. *Gut Microbes*. 2024;16(1):2333748. doi: 10.1080/19490976.2024.2333748
65. Huttner BD, de Lastours V, Wassenberg M, et al. A 5-day course of oral antibiotics followed by faecal transplantation to eradicate carriage of multidrug-resistant Enterobacteriaceae: a randomized clinical trial. *Clin Microbiol Infect*. 2019;25(7):830–838. doi: 10.1016/j.cmi.2018.12.009
66. Hyun J, Lee SK, Cheon JH, et al. Faecal microbiota transplantation reduces amounts of antibiotic resistance genes in patients with multidrug-resistant organisms. *Antimicrob Resist Infect Control*. 2022;11(1):20. doi: 10.1186/s13756-022-01064-4
67. Alonso CD, Treadway SB, Hanna DB, et al. Epidemiology and outcomes of clostridium difficile infections in hematopoietic stem cell transplant recipients. *Clin Infect Dis*. 2012;54(8):1053–1063. doi: 10.1093/cid/cir1035
68. Bhutani D, Jaiyeoba C, Kim S, et al. Relationship between clostridium difficile infection and gastrointestinal graft versus host disease in recipients of allogeneic stem cell transplantation. *Bone Marrow Transplant*. 2019;54(1):164–167. doi: 10.1038/s41409-018-0270-x
69. Bruminhent J, Wang Z-X, Hu C, et al. Clostridium difficile colonization and disease in patients undergoing hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant*. 2014;20(9):1329–1334. doi: 10.1016/j.bbmt.2014.04.026
70. Dubberke ER, Reske KA, Olsen MA, et al. Epidemiology and outcomes of Clostridium difficile infection in allogeneic hematopoietic cell and lung transplant recipients [article]. *Transpl Infect Disease*. 2018;20(2). doi: 10.1111/tid.12855
71. Re J, El Atrouni W, Shune L, et al. Clostridioides difficile infection and risk of acute graft-versus-host disease among allogeneic hematopoietic stem cell transplantation recipients. *Transplant And Cellular Ther*. 2021;27(2): e176. 1–. e176. 8. doi: 10.1016/j.jtct.2020.10.009
72. Kamboj M, Xiao K, Kaltsas A, et al. Clostridium difficile infection after allogeneic hematopoietic stem cell transplant: Strain diversity and outcomes associated with NAP1/027. *Biol Blood Marrow Transplant*. 2014;20(10):1626–1633. doi: 10.1016/j.bbmt.2014.06.025
73. Kinnebrew MA, Lee YJ, Jenq RR, et al. Early clostridium difficile infection during allogeneic hematopoietic stem cell transplantation. *PLOS ONE*. 2014;9(3):e90158. doi: 10.1371/journal.pone.0090158
74. Willems L, Porcher R, Lafaurie M, et al. Clostridium difficile infection after allogeneic hematopoietic stem cell transplantation: incidence, risk factors, and outcome. *Biol Blood Marrow Transplant*. 2012;18(8):1295–1301. doi: 10.1016/j.bbmt.2012.02.010
75. Trifilio SM, Pi J, Mehta J. Changing epidemiology of clostridium difficile-associated disease during stem cell transplantation. *Biol Blood Marrow Transplant*. 2013;19(3):405–409. doi: 10.1016/j.bbmt.2012.10.030
76. Patil R, Blankenship L. Proton pump inhibitors and clostridium difficile infection: are we propagating an already rapidly growing healthcare problem? *Gastroenterology Res*. 2013;6(5):171. doi: 10.4021/gr575w
77. Giuliano M, Barza M, Jacobus N, et al. Effect of broad-spectrum parenteral antibiotics on composition of intestinal microflora of humans. *Antimicrob Agents Chemother*. 1987;31(2):202–206. doi: 10.1128/AAC.31.2.202
78. Sörgel F, Kinzig M. The chemistry, pharmacokinetics and tissue distribution of piperacillin/tazobactam. *J Antimicrob Chemother*. 1993;31(suppl_A):39–60. doi: 10.1093/jac/31.suppl_A.39
79. Thabit AK. Antibiotics in the biliary tract: a review of the pharmacokinetics and clinical outcomes of antibiotics penetrating the bile and gallbladder wall. *Pharmacother*. 2020;40(7):672–691. doi: 10.1002/phar.2431
80. Degraeve AL, Haufroid V, Liorot A, et al. Gut microbiome modulates tacrolimus pharmacokinetics through the transcriptional regulation of ABCB1. *Microbiome*. 2023;11(1):138. doi: 10.1186/s40168-023-01578-y
81. Saqr A, Carlson B, Staley C, et al. Reduced enterohepatic recirculation of mycophenolate and lower blood concentrations are associated with the stool bacterial microbiome after hematopoietic cell transplantation. *Transplant Cellular Ther*. 2022;28(7): e372. 1–. e372. 9. doi: 10.1016/j.jtct.2022.04.018
82. Barreto JN, Aitken SL, Krantz EM, et al. Variation in clinical practice and attitudes on antibacterial management of fever and neutropenia in patients with hematologic malignancy: a survey of cancer centers across the United States. *Open Forum Infect Dis*. 2022;9(3). doi: 10.1093/ofid/ofac005
83. Verlinden A, Mikulska M, Knelange NS, et al. Current antimicrobial practice in febrile neutropenia across Europe and Asia: the EBMT infectious disease working party survey. *Bone Marrow Transplant*. 2020;55(8):1588–1594. doi: 10.1038/s41409-020-0811-y
84. Aguilar-Guisado M, Espigado I, Martín-Peña A, et al. Optimisation of empirical antimicrobial therapy in patients with haematological

- malignancies and febrile neutropenia (how long study): an open-label, randomised, controlled phase 4 trial. *Lancet Haematol.* 2017;4(12):e573–e583. doi: [10.1016/S2352-3026\(17\)30211-9](https://doi.org/10.1016/S2352-3026(17)30211-9)
- A phase 4 study demonstrated safe discontinuation of empiric antibiotic after 72 hours in selected patients.**
85. Snyder M, Pasikhova Y, Baluch A. Early antimicrobial de-escalation and stewardship in adult hematopoietic stem cell transplantation recipients: retrospective review. In: *Open forum infectious diseases*. vol. 4. US: Oxford University Press; 2017. p. ofx226. doi: [10.1093/ofid/ofx226](https://doi.org/10.1093/ofid/ofx226)
 86. Abbott R, Keller J, Slade M, et al. Severe cytokine-release syndrome after T cell-replete peripheral Blood haploidentical donor transplantation is associated with poor survival and anti-IL-6 therapy is safe and well tolerated. *Biol Of Blood And Marrow Transplant.* 2016;22(10):1851–1860. doi: [10.1016/j.bbmt.2016.06.010](https://doi.org/10.1016/j.bbmt.2016.06.010)
 87. van der Velden WJ, Herbers AH, Netea MG, et al. Mucosal barrier injury, fever and infection in neutropenic patients with cancer: introducing the paradigm febrile mucositis. *Br J Haematol.* 2014;167(4):441–452. doi: [10.1111/bjh.13113](https://doi.org/10.1111/bjh.13113)
 88. Feng S, Rao G, Wei X, et al. Clinical metagenomic sequencing of plasma microbial cell-free DNA for febrile neutropenia in patients with acute leukaemia. *Clin Microbiol Infect.* 2024;30(1):107–113. doi: [10.1016/j.cmi.2023.05.034](https://doi.org/10.1016/j.cmi.2023.05.034)
 89. Kokai-Kun JF, Le C, Trout K, et al. Ribaxamase, an orally administered β -lactamase, diminishes changes to acquired antimicrobial resistance of the gut resistome in patients treated with ceftriaxone. *IDR.* 2020;2020:2521–2535. doi: [10.2147/IDR.S260258](https://doi.org/10.2147/IDR.S260258)
 90. de Gunzburg J, Ducher A, Modess C, et al. Targeted adsorption of molecules in the colon with the novel adsorbent-based medicinal product, DAV132: a proof of concept study in healthy subjects. *J Clin Pharmacol.* 2015;55(1):10–16. doi: [10.1002/jcph.359](https://doi.org/10.1002/jcph.359)
 91. de Gunzburg J, Ghoslane A, Ducher A, et al. Protection of the human gut microbiome from antibiotics. *J Infect Dis.* 2018;217(4):628–636. doi: [10.1093/infdis/jix604](https://doi.org/10.1093/infdis/jix604)
 92. Guk J, Guedj J, Burdet C, et al. Modeling the effect of DAV132, a novel colon-targeted adsorbent, on fecal concentrations of moxifloxacin and gut microbiota diversity in healthy volunteers. *Clin Pharmacol Ther.* 2021;109(4):1045–1054. doi: [10.1002/cpt.1977](https://doi.org/10.1002/cpt.1977)
 93. Ramphal R. Changes in the etiology of bacteremia in febrile neutropenic patients and the susceptibilities of the currently isolated pathogens. *Clin Infect Dis.* 2004;39(Supplement_1):S25–S31. doi: [10.1086/383048](https://doi.org/10.1086/383048)
 94. Beyar-Katz O, Dickstein Y, Borok S, et al. Empirical antibiotics targeting gram-positive bacteria for the treatment of febrile neutropenic patients with cancer. *Cochrane Database Syst Rev.* 2017;6(6):Cd003914. doi: [10.1002/14651858.CD003914.pub4](https://doi.org/10.1002/14651858.CD003914.pub4)
 95. Paul M, Borok S, Fraser A, et al. Additional anti-gram-positive antibiotic treatment for febrile neutropenic cancer patients. *Cochrane Database Syst Rev.* 2005;2005(3):Cd003914. doi: [10.1002/14651858.CD003914.pub2](https://doi.org/10.1002/14651858.CD003914.pub2)
 96. Cometta A, Kern WV, De Bock R, et al. Vancomycin versus placebo for treating persistent fever in patients with neutropenic cancer receiving piperacillin-tazobactam monotherapy. *Clin Infect Dis.* 2003;37(3):382–389. doi: [10.1086/376637](https://doi.org/10.1086/376637)
 97. Guarana M, Nucci M, Nouér SA. Shock and early death in hematologic patients with febrile neutropenia. *Antimicrob Agents Chemother.* 2019;63(11). doi: [10.1128/AAC.01250-19](https://doi.org/10.1128/AAC.01250-19)
 98. Freifeld AG, Bow EJ, Sepkowitz KA, et al. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 update by the infectious diseases Society of America. *Clin Infect Dis.* 2011;52(4):427–431. doi: [10.1093/cid/ciq147](https://doi.org/10.1093/cid/ciq147)
 99. Black E, Lau TT, Ensom MH. Vancomycin-induced neutropenia: is it dose- or duration-related? *Ann Pharmacother.* 2011;45(5):629–638. doi: [10.1345/aph.1P583](https://doi.org/10.1345/aph.1P583)
 100. Geraci J. Some laboratory and clinical experiences with a new antibiotic, vancomycin. *Proc Staff Meet Mayo Clin.* 1956;31:564–582.
 101. Currie BP, Lemos-Filho L. Evidence for biliary excretion of vancomycin into stool during intravenous therapy: potential implications for rectal colonization with vancomycin-resistant enterococci. *Antimicrob Agents Chemother.* 2004;48(11):4427–4429. doi: [10.1128/AAC.48.11.4427-4429.2004](https://doi.org/10.1128/AAC.48.11.4427-4429.2004)
 102. Eubank TA, Hu C, Gonzales-Luna AJ, et al. Detectable vancomycin stool concentrations in hospitalized patients with diarrhea given intravenous vancomycin. *Pharmacoeconomics.* 2023;2(4):283–288. doi: [10.3390/pharma2040024](https://doi.org/10.3390/pharma2040024)
 103. Chanderraj R, Millar JA, Patel TS, et al. Vancomycin-resistant enterococcus acquisition in a tertiary care Hospital: testing the roles of antibiotic use, Proton pump inhibitor use, and colonization pressure. *Open Forum Infect Dis.* 2019;6(4):ofz139. doi: [10.1093/ofid/ofz139](https://doi.org/10.1093/ofid/ofz139)
 104. Zaas AK, Song X, Tucker P, et al. Risk factors for development of vancomycin-resistant enterococcal bloodstream infection in patients with cancer who are colonized with vancomycin-resistant enterococci. *Clin Infect Dis.* 2002;35(10):1139–1146. doi: [10.1086/342904](https://doi.org/10.1086/342904)
 105. Zacharioudakis IM, Zervou FN, Ziakas PD, et al. Vancomycin-resistant enterococci colonization among dialysis patients: a meta-analysis of prevalence, risk factors, and significance. *Am J Kidney Dis.* 2015;65(1):88–97. doi: [10.1053/j.ajkd.2014.05.016](https://doi.org/10.1053/j.ajkd.2014.05.016)
 106. Kamboj M, Cohen N, Huang YT, et al. Impact of empiric treatment for vancomycin-resistant enterococcus in colonized patients early after allogeneic hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant.* 2019;25(3):594–598. doi: [10.1016/j.bbmt.2018.11.008](https://doi.org/10.1016/j.bbmt.2018.11.008)
 107. Karp JE, Merz WG, Hendricksen C, et al. Oral norfloxacin for prevention of gram-negative bacterial infections in patients with acute leukemia and granulocytopenia: a randomized, double-blind, placebo-controlled trial. *Ann Intern Med.* 1987;106(1):1–7. doi: [10.7326/0003-4819-106-1-1](https://doi.org/10.7326/0003-4819-106-1-1)
 108. Levine AS, Siegel SE, Schreiber AD, et al. Protected environments and prophylactic antibiotics: a prospective controlled study of their utility in the therapy of acute leukemia. *N Engl J Med.* 1973;288(10):477–483. doi: [10.1056/NEJM197303082881001](https://doi.org/10.1056/NEJM197303082881001)
 109. Gafter-Gvili A, Fraser A, Paul M, et al. Antibiotic prophylaxis for bacterial infections in afebrile neutropenic patients following chemotherapy. *Cochrane Database Systematic Rev.* 2012;2012(1). doi: [10.1002/14651858.CD004386.pub3](https://doi.org/10.1002/14651858.CD004386.pub3)
 110. Lehrnbecher T, Fisher BT, Phillips B, et al. Guideline for antibacterial prophylaxis administration in pediatric cancer and hematopoietic stem cell transplantation. *Clin Infect Dis.* 2020;71(1):226–236. doi: [10.1093/cid/ciz1082](https://doi.org/10.1093/cid/ciz1082)
 111. Taplitz RA, Kennedy EB, Bow EJ, et al. Antimicrobial prophylaxis for adult patients with cancer-related immunosuppression: ASCO and IDSA clinical practice guideline update. *J Clin Oncol.* 2018;36(30):3043–3054. doi: [10.1200/JCO.18.00374](https://doi.org/10.1200/JCO.18.00374)
 112. Classen AY, Henze L, von Lilienfeld-Toal M, et al. Primary prophylaxis of bacterial infections and pneumocystis jirovecii pneumonia in patients with hematologic malignancies and solid tumors: 2020 updated guidelines of the infectious diseases working party of the German Society of hematology and Medical Oncology (AGIHO/DGHO). *Ann Hematol.* 2021;100(6):1603–1620. doi: [10.1007/s00277-021-04452-9](https://doi.org/10.1007/s00277-021-04452-9)
 113. Egan G, Robinson PD, Martinez JP, et al. Efficacy of antibiotic prophylaxis in patients with cancer and hematopoietic stem cell transplantation recipients: a systematic review of randomized trials. *Cancer Med.* 2019;8(10):4536–4546. doi: [10.1002/cam4.2395](https://doi.org/10.1002/cam4.2395)
 114. Kimura S, Akahoshi Y, Nakano H, et al. Antibiotic prophylaxis in hematopoietic stem cell transplantation. A meta-analysis of randomized controlled trials. *J Infect.* 2014;69(1):13–25. doi: [10.1016/j.jinf.2014.02.013](https://doi.org/10.1016/j.jinf.2014.02.013)
 115. Mikulska M, Averbuch D, Tissot F, et al. Fluoroquinolone prophylaxis in hematological cancer patients with neutropenia: ECIL critical appraisal of previous guidelines. *J Infect.* 2018;76(1):20–37. doi: [10.1016/j.jinf.2017.10.009](https://doi.org/10.1016/j.jinf.2017.10.009)
 116. Owattanapanich W, Chayakulkeeree M. Efficacy of levofloxacin as an antibacterial prophylaxis for acute leukemia patients receiving intensive chemotherapy: a systematic review and meta-analysis. *Hematology.* 2019;24(1):362–368. doi: [10.1080/16078454.2019.1589706](https://doi.org/10.1080/16078454.2019.1589706)

117. Hoffman T, Atamna A, Litchevsky V, et al. Fluoroquinolone prophylaxis during conventional chemotherapy or hematopoietic stem cell transplantation for acute leukemia - pros and cons. *Acta Haematol.* **2024**;147(2):186–197. doi: [10.1159/000535119](https://doi.org/10.1159/000535119)
118. Alexander S, Fisher BT, Gaur AH, et al. Effect of levofloxacin prophylaxis on bacteremia in children with acute leukemia or undergoing hematopoietic stem cell transplantation: a randomized clinical trial. *JAMA.* **2018**;320(10):995–1004. doi: [10.1001/jama.2018.12512](https://doi.org/10.1001/jama.2018.12512)
119. Kern WV, Weber S, Dettenkofer M, et al. Impact of fluoroquinolone prophylaxis during neutropenia on bloodstream infection: data from a surveillance program in 8755 patients receiving high-dose chemotherapy for haematologic malignancies between 2009 and 2014. *J Infect.* **2018**;77(1):68–74. doi: [10.1016/j.jinf.2018.05.004](https://doi.org/10.1016/j.jinf.2018.05.004)
120. Gardner JC, Courter JD, Dandoy CE, et al. Safety and efficacy of prophylactic levofloxacin in pediatric and adult hematopoietic stem cell transplantation patients. *Transpl Cell Ther.* **2022**;28(3):e167.1–e167.5. doi: [10.1016/j.jtct.2021.11.017](https://doi.org/10.1016/j.jtct.2021.11.017)
121. Stern A, Henig I, Cohen M, et al. Impact of discontinuing routine fluoroquinolone prophylaxis in neutropenic allogeneic haematopoietic stem cell transplant recipients: an observational study. *J Antimicrob Chemother.* **2024**;79(12):3289–3296. doi: [10.1093/jac/dkac360](https://doi.org/10.1093/jac/dkac360)
122. Clerici D, Galli L, Greco R, et al. Levofloxacin prophylaxis vs no prophylaxis in patients with neutropenia within an endemic country for carbapenem-resistant GNB. *Blood Adv.* **2023**;7(9):1621–1634. doi: [10.1182/bloodadvances.2022008226](https://doi.org/10.1182/bloodadvances.2022008226)
- **Fluoroquinolone use may associated with MDR pathogens.**
123. Verlinden A, Schroyens WA, Gadisseur AP. Clinical and microbiological impact of long-term discontinuation of fluoroquinolone prophylaxis in haematological patients with prolonged profound neutropenia. *Eur J Haematol.* **2021**;107(3):377–379. doi: [10.1111/ejh.13670](https://doi.org/10.1111/ejh.13670)
124. Satlin MJ, Chavda KD, Baker TM, et al. Colonization with levofloxacin-resistant extended-spectrum β -lactamase-producing Enterobacteriaceae and risk of bacteremia in hematopoietic stem cell transplant recipients. *Clin Infect Dis.* **2018**;67(11):1720–1728. doi: [10.1093/cid/ciy363](https://doi.org/10.1093/cid/ciy363)
125. Satlin MJ, Chen L, Douglass C, et al. Colonization with fluoroquinolone-resistant Enterobacterales decreases the effectiveness of fluoroquinolone prophylaxis in hematopoietic cell transplant recipients. *Clin Infect Dis.* **2021**;73(7):1257–1265. doi: [10.1093/cid/ciab404](https://doi.org/10.1093/cid/ciab404)
126. Slavin M, Lingaratnam S, Mileschkin L, et al. Use of antibacterial prophylaxis for patients with neutropenia. *Intern Med J.* **2011**;41(1b):102–109. doi: [10.1111/j.1445-5994.2010.02341.x](https://doi.org/10.1111/j.1445-5994.2010.02341.x)
127. DWv B, Roodenburg J, Heidt PJ, et al. Mitigation of secondary disease of allogeneic mouse radiation chimeras by modification of the intestinal microflora. *JNCI: J Natl Cancer Inst.* **1974**;52(2):401–404. doi: [10.1093/jnci/52.2.401](https://doi.org/10.1093/jnci/52.2.401)
128. Jones JM, Wilson R, Bealmeier PM. Mortality and gross pathology of secondary disease in germfree mouse radiation chimeras. *Radiat Res.* **1971**;45(3):577–588. doi: [10.2307/3573066](https://doi.org/10.2307/3573066)
129. Vossen J, Heidt P, Van Den Berg H, et al. Prevention of infection and graft-versus-host disease by suppression of intestinal microflora in children treated with allogeneic bone marrow transplantation. *Eur J Clin Microbiol Infect Dis.* **1990**;9(1):14–23. doi: [10.1007/BF01969527](https://doi.org/10.1007/BF01969527)
130. Vossen JM, Guiot HF, Lankester AC, et al. Complete suppression of the gut microbiome prevents acute graft-versus-host disease following allogeneic bone marrow transplantation. *PLOS ONE.* **2014**;9(9):e105706. doi: [10.1371/journal.pone.0105706](https://doi.org/10.1371/journal.pone.0105706)
131. Beelen DW, Haralambie E, Brandt H, et al. Evidence that sustained growth suppression of intestinal anaerobic bacteria reduces the risk of acute graft-versus-host disease after sibling marrow transplantation. *Blood.* **1992**;80(10):2668–2676. doi: [10.1182/blood.V80.10.2668.bloodjournal80102668](https://doi.org/10.1182/blood.V80.10.2668.bloodjournal80102668)
132. Beelen DW, Elmaagacli A, Müller K-D, et al. Influence of intestinal bacterial decontamination using metronidazole and ciprofloxacin or ciprofloxacin alone on the development of acute graft-versus-host disease after Marrow transplantation in patients with hematologic malignancies: final results and long-term follow-up of an open-label prospective randomized trial. *Blood.* **1999**;93(10):3267–3275.
- **A key study showing gut decontamination reduces GVHD.**
133. Wlodarska M, Kostic AD, Xavier RJ. An integrative view of microbiome-host interactions in inflammatory bowel diseases. *Cell Host Microbe.* **2015**;17(5):577–591. doi: [10.1016/j.chom.2015.04.008](https://doi.org/10.1016/j.chom.2015.04.008)
134. Gavrilaki E, Mallouri D, Laspa E, et al. Open-label randomized controlled study of ciprofloxacin vs Rifaximin as neutropenia prophylaxis in allogeneic hematopoietic stem cell transplantation. *Transplant Proc.* **2024**;56(2):380–385. doi: [10.1016/j.transproceed.2023.12.010](https://doi.org/10.1016/j.transproceed.2023.12.010)
135. Daoud-Asfour H, Henig I, Ghersin I, et al. Omitting ciprofloxacin prophylaxis in patients undergoing allogeneic hematopoietic stem cell transplantation and its impact on clinical outcomes and microbiome structure [article]. *Transplant Cellular Ther.* **2022**;28(3):e168.1–e168.8. doi: [10.1016/j.jtct.2021.12.012](https://doi.org/10.1016/j.jtct.2021.12.012)
136. Rashid M-U, Zaura E, Buijs MJ, et al. Determining the long-term effect of antibiotic administration on the human normal intestinal microbiota using culture and pyrosequencing methods. *Clin Infect Dis.* **2015**;60(suppl_2):S77–S84. doi: [10.1093/cid/civ137](https://doi.org/10.1093/cid/civ137)
137. Ziegler M, Han JH, Landsburg D, et al. Impact of levofloxacin for the prophylaxis of bloodstream infection on the gut microbiome in patients with hematologic malignancy [review]. *Open Forum Infect Dis.* **2019**;6(7). doi: [10.1093/ofid/ofz252](https://doi.org/10.1093/ofid/ofz252)
138. Rashidi A, Kaiser T, Holtan SG, et al. Levaquin Gets a Pass. *Biol Of Blood And Marrow Transplant.* **2020**;26(4):778–781. doi: [10.1016/j.bbmt.2019.12.722](https://doi.org/10.1016/j.bbmt.2019.12.722)
139. Hakki M, Humphries RM, Hemarajata P, et al. Fluoroquinolone prophylaxis selects for meropenem-nonsusceptible pseudomonas aeruginosa in patients with hematologic malignancies and hematopoietic cell transplant recipients. *Clin Infect Dis.* **2018**;68(12):2045–2052. doi: [10.1093/cid/ciy825](https://doi.org/10.1093/cid/ciy825)
140. Leardini D, Gambuti G, Muratore E, et al. Levofloxacin prophylaxis in pediatric and young adult allogeneic hematopoietic stem cell transplantation recipients does not prevent infective complications and infections-related deaths. *Open Forum Infect Dis.* **2024**;12(2). doi: [10.1093/ofid/ofae707](https://doi.org/10.1093/ofid/ofae707)
141. Zimmer AJ, Stohs E, Meza J, et al. Bloodstream infections in hematologic malignancy patients with fever and neutropenia: are empirical antibiotic therapies in the United States still effective? *Open Forum Infect Dis.* **2022**;9(7). doi: [10.1093/ofid/ofac240](https://doi.org/10.1093/ofid/ofac240)
142. Averbuch D, Tridello G, Hoek J, et al. Antimicrobial resistance in gram-negative rods causing bacteremia in hematopoietic stem cell transplant recipients: intercontinental prospective study of the infectious diseases working party of the European bone marrow transplantation group. *Clin Infect Dis.* **2017**;65(11):1819–1828. doi: [10.1093/cid/cix646](https://doi.org/10.1093/cid/cix646)
143. Slimings C, Riley TV. Antibiotics and healthcare facility-associated clostridioides difficile infection: systematic review and meta-analysis 2020 update. *J Antimicrob Chemother.* **2021**;76(7):1676–1688. doi: [10.1093/jac/dkab091](https://doi.org/10.1093/jac/dkab091)
144. Neuerburg CKF, Schmitz F, Schmitz M-T, et al. Antibiotic prophylaxis during allogeneic stem cell transplantation—A comprehensive single center retrospective analysis. *Transplant Cellular Ther.* **2024**;30(12):e1195.1–e1195.13. doi: [10.1016/j.jtct.2024.09.011](https://doi.org/10.1016/j.jtct.2024.09.011)
145. Signorelli J, Zimmer A, Liewer S, et al. Incidence of febrile neutropenia in autologous hematopoietic stem cell transplant (HSCT) recipients on levofloxacin prophylaxis. *Transpl Infect Disease.* **2020**;22(2):e13225. doi: [10.1111/tid.13225](https://doi.org/10.1111/tid.13225)